Bag of Words

- In this model, a text (such as a sentence or a document) is represented as the bag (multiset) of its words, disregarding grammar and even word order but keeping multiplicity.
- A document is simply the collection of terms/words/lexemes with frequencies.
- Similarly a query is also transformed as BoW representation.

Bag of Words

- Example

  D1: {John likes to watch movies. Mary likes movies too. }

  D2: {John also likes to watch football games.}

  D3: { Mary does not like football games }
- Bow Representation

  D1: {John, like,watch,movie,Mary,like,movie}

  D2: {John,like,watch,football,game}

  D3: { Mary,like,football,game }
- Query Term

  Q: {like football game}

Bag of Words

- Scores

  Common Terms in document and query over total terms

  (D1,Q) = (D1 ∩ Q) / (D1 U Q) = 1/8

  (D2,Q)= 3/5

  (D3,Q)= 3/4

- Issues

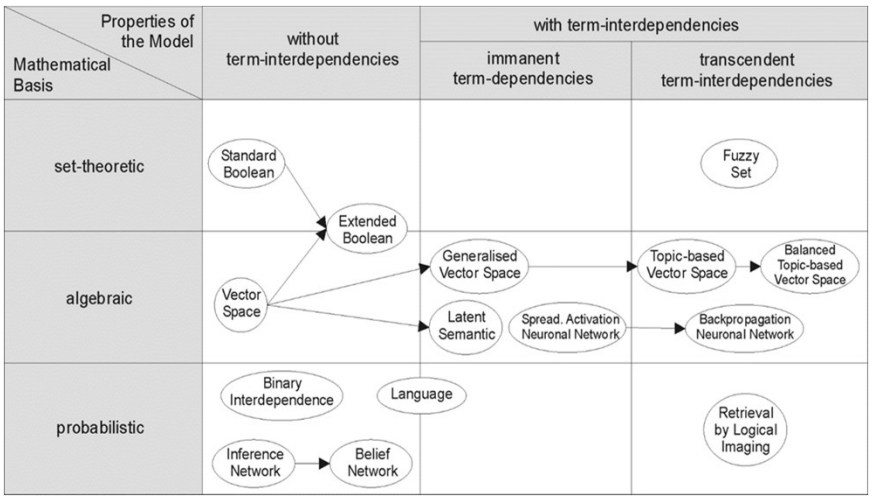
  Document size affects the overall similarity.

  Frequency of common terms are ignored.

  Similarity is only based on common terms, which are treated as independent of each other.

  Vector Space Model is the answer to these problems.

IR Models



Vector Space Model

- Vector Space Model is an algebraic model for representing text documents (and any objects, in general) as vectors of identifiers, such as, for example, terms of a document.
- It is used in information filtering, information retrieval, indexing and relevancy rankings.
- Its first use was in the SMART Information Retrieval System.

Vector Space Model

- Assumptions
  - Document and Queries are represented in the vector space
  - Relevant documents are close to the query in term of distance in vector space
  - All terms are independent
  - Term weighting can play a role in distance function and importance

Vector Representation of Documents

- From offline collection build a dictionary.
- Let the size of vocabulary be “V”
- So a V-dimensional vector is used for representing documents in limited vocabulary space.
- This dictionary is kept in alphabetical order of vocabulary

  d = <0,1,0,1,….0>

Vector Representation of Query

- The query is also represented similar to document.
- A V-dimensional vector is also used for query, only the term present in query are non-zero vector for the query.

Cosine Similarity

- One of the limitation of Boolean Model is the similarity measure it offers.
- In vector space model, similarity is calculated by cosine measure between query and document.

  $\cos \Theta = (d \cdot q) / (|d| \cdot |q|)$

Term Frequency

- In initial vector space model, we only use a term present for the v-dimensional calculation.
- What if a term appears more frequent in a document?
- Can we use term-frequency
  - Term Frequency – how many times a term appear in a document.
- Problem with frequent term?

Sub linear " tf " Scaling

- Term Frequency can be scaled in different ways (tf)
- Log Linear $1 + \log(\mathrm{tf}_{t,d})$
- Augmented $0.5 + \frac{0.5 \times \mathrm{tf}_{t,d}}{\max_t(\mathrm{tf}_{t,d})}$
- Maximum Frequency normalization

Document Frequency

- Document Frequency – How many documents contains a given term? (df)?
- Scaling the df $\mathrm{idf}_t = \log \frac{N}{\mathrm{df}_t}.$
- Augmented $\max\{0, \log \frac{N - \mathrm{df}_t}{\mathrm{df}_t}\}$

Weighting Schemes

- One of the popular and effective weighting scheme is
  - term frequency * inverse document frequency

$$\text{tf-idf}_{t,d} = \text{tf}_{t,d} \times \text{idf}_t.$$

  - In other words, tf-idft, assigns to term t a weight in document d that is
    - Highest when t occurs many times within a small number of documents (thus lending high discriminating power to those documents);
    - Lower when the term occurs fewer times in a document, or occurs in many documents (thus offering a less pronounced relevance signal);
    - Lowest when the term occurs in virtually all documents.

Different Variants

Term frequency		Document frequency		Normalization	
n (natural)	$\text{tf}_{t,d}$	n (no)	1	n (none)	1
l (logarithm)	$1+\log(\text{tf}_{t,d})$	t (idf)	$\log \frac{N}{\text{df}_t}$	c (cosine)	$\frac{1}{\sqrt{w_1^2+w_2^2+\ldots+w_M^2}}$
a (augmented)	$0.5+\frac{0.5\times\text{tf}_{t,d}}{\max_t(\text{tf}_{t,d})}$	p (prob idf)	$\max\{0, \log\frac{N-\text{df}_t}{\text{df}_t}\}$	u (pivoted unique)	$1/u$ (Section 6.4.4)
b (boolean)	$\begin{cases}1 & \text{if } \text{tf}_{t,d}>0\\ 0 & \text{otherwise}\end{cases}$			b (byte size)	$1/CharLength^{\alpha}, \alpha<1$
L (log ave)	$\frac{1+\log(\text{tf}_{t,d})}{1+\log(\text{ave}_{t\in d}(\text{tf}_{t,d}))}$				

▶ **Figure 6.15** SMART notation for tf-idf variants. Here *CharLength* is the number of characters in the document.

Advantages of VSM

- Simple model based on linear algebra
- Term weights not binary
- Allows computing a continuous degree of similarity between queries and documents
- Allows ranking documents according to their possible relevance
- Allows partial matching
- Last 20 years of working knowledge

Disadvantages of VSM

- The order in which the terms appear in the document is lost in the vector space representation.
- Theoretically assumes terms are statistically independent.
- Search keywords must precisely match document terms; word substrings might result in a "false positive match“
- Semantic sensitivity; documents with similar context but different term vocabulary won't be associated, resulting in a "false negative match".

SMART System

- The SMART system did a rigor research on different schemes, for document and query they have used mnemonic ddd.qqq
- The first letter in each triplet specifies the term frequency component of the weighting, the second the document frequency component, and the third the form of normalization used.
- For example, a very standard weighting scheme is lnc.ltc, where the document vector has log-weighted term frequency, no idf (for both effectiveness and efficiency reasons), and cosine normalization, while the query vector uses log-weighted term frequency, idf weighting, and cosine normalization.